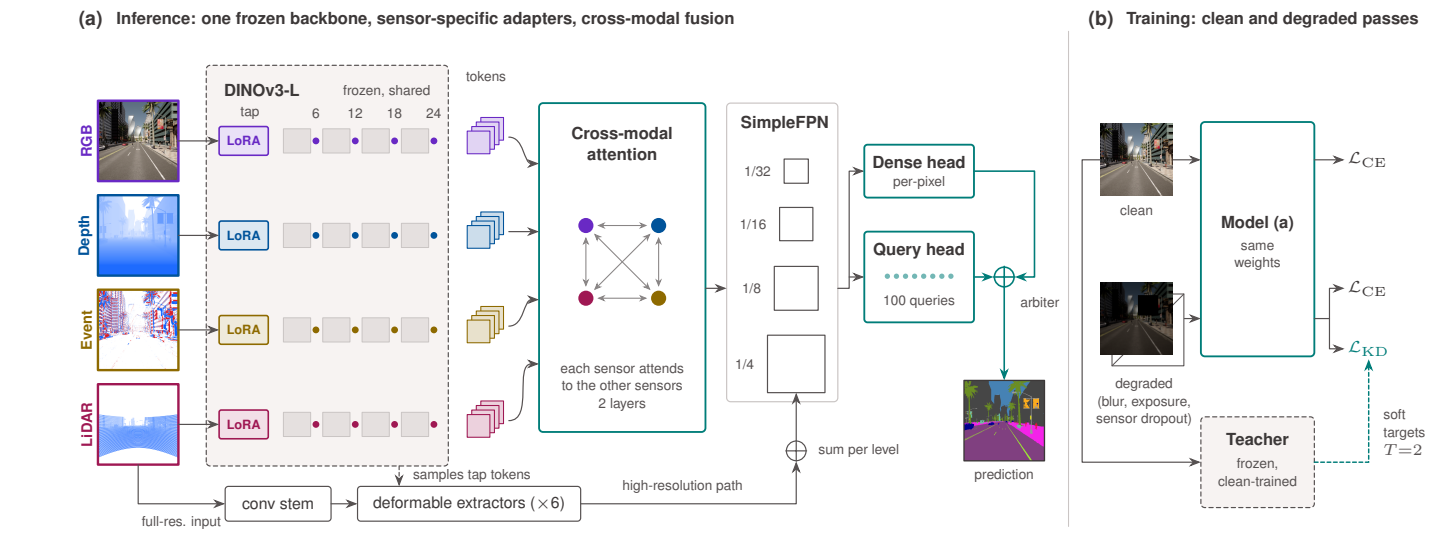

(a) Inference: one frozen backbone, sensor-specific adapters, cross-modal fusion
DINOv3-L
frozen, shared
tap
6
12
18
24
RGB
Depth
Event
LiDAR
LoRA
LoRA
LoRA
LoRA
tokens
Cross-modal attention
each sensor attends to the other sensors
2 layers
SimpleFPN
1/32
1/16
1/8
1/4
Dense head
per-pixel
Query head
100 queries
arbiter
prediction
sum per level
samples tap tokens
conv stem
deformable extractors (×6)
high-resolution path
full-res. input
(b) Training: clean and degraded passes
clean
degraded
(blur, exposure,
sensor dropout)
Model (a)
same
weights
$\mathcal{L}_{\mathrm{CE}}$
$\mathcal{L}_{\mathrm{CE}}$
$\mathcal{L}_{\mathrm{KD}}$
Teacher
frozen,
clean-trained
soft
targets
$T=2$